

An Examination of Adverse Childhood Experiences as a Predictor of Physical Activity in Young Adults

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Methods

Power analyses indicated that a sample size of 328 participants would be needed for mediation analyses, at 80% power. **388 undergraduates** ($M_{age} = 20.33$, $SD = 4.07$) completed an online survey in return for course credit. Participants were retained for analysis if they reported their age, sex, household income, and completed at least one of the main measures ($N = 357$). 58.1% indicated female sex and 94.4% identified as Hispanic. ACEs were measured using the 10-item **Adverse Childhood Experiences Questionnaire** (ACE-Q), PA was measured using the 7-item **International Physical Activity Questionnaire**, short form (IPAQ-SF) and perceived barriers/benefits to exercise were measured using the subscales from the 43-item **Exercise Benefits and Barriers Scale** (EBBS). Statistical analyses were performed using SPSS 28 with 5,000 bootstrapped samples. Correlation was calculated using Spearman's r . A parallel mediation analysis from ACEs to PA through perceived exercise benefits/barriers was conducted using Hayes' PROCESS Macro V4.2, controlling for age, sex, and annual family household income.

Discussion

Based on a bivariate nonparametric correlation, our results indicate that fewer ACEs are associated with greater PA. Although this association does not hold when other variables are controlled for, ACEs was found to indirectly affect PA through perceived benefits (but not perceived barriers). These findings build on the limited existing evidence regarding the ACEs-PA association and mediators of this association. Future research would benefit from better measures. For example, one of the most commonly indicated barriers was the item "exercise tires me". It is not clear that agreeing to that item would necessarily be perceived as a barrier, however, it could just be perceived as a normal healthy response to exercise. Additionally, ACEs relies on retrospective self-report and may be subject to bias. More objective measures of PA would also be important for future research, based on approaches such as actigraphy.

Introduction

Adverse childhood experiences are prevalent, with more than two-thirds (71%) of young adults reporting having experienced one or more ACEs and 22.1% reporting four or more. Amidst a broad interest in the association between ACEs and health, researchers have recently begun to examine the association between ACEs and physical activity (PA). Although PA is considered an important health-promoting behavior, the proportion of adults engaging in recommended levels of PA is alarmingly low. A limited number of studies examining the ACEs-PA relationship exist and the findings have been inconsistent. This examines the relationship in college students and additionally examines perceived barriers and benefits to exercise as potential mediators.

We hypothesized that (i) The number of ACEs will be negatively associated with PA and (ii) the association between ACEs and PA will be mediated by perceived barriers and benefits to exercise.

Results

Based on Spearman's correlation, ACEs was found to be negatively correlated with PA ($r_s(355) = -.122$, $p = .011$). However, when controlling for other variables, the mediation analysis indicated that the direct effect of ACEs to PA was not statistically significant. Despite no direct effect of ACEs on PA, the parallel mediation analysis indicated a significant indirect effect via perceived benefits (unstandardized indirect effect = -60.1489, 95% CI [-123.2647, -4.1576]) but not through perceived barriers (unstandardized indirect effect = -13.7802, 95% CI [-49.2401, 13.2774]. Mediation model paths and statistics are shown in Figure 1.

The top 3 barriers for students were time commitment, fatigue, and accessibility issues, and the top 3 benefits were muscle strength, personal accomplishment, and cardiovascular health. Participants also reported other benefits not addressed in the EBBS, including immediate mental health relief, long-term physical health gains, enhanced self-esteem, social engagement, lifestyle improvements, and prospects of healthy aging.

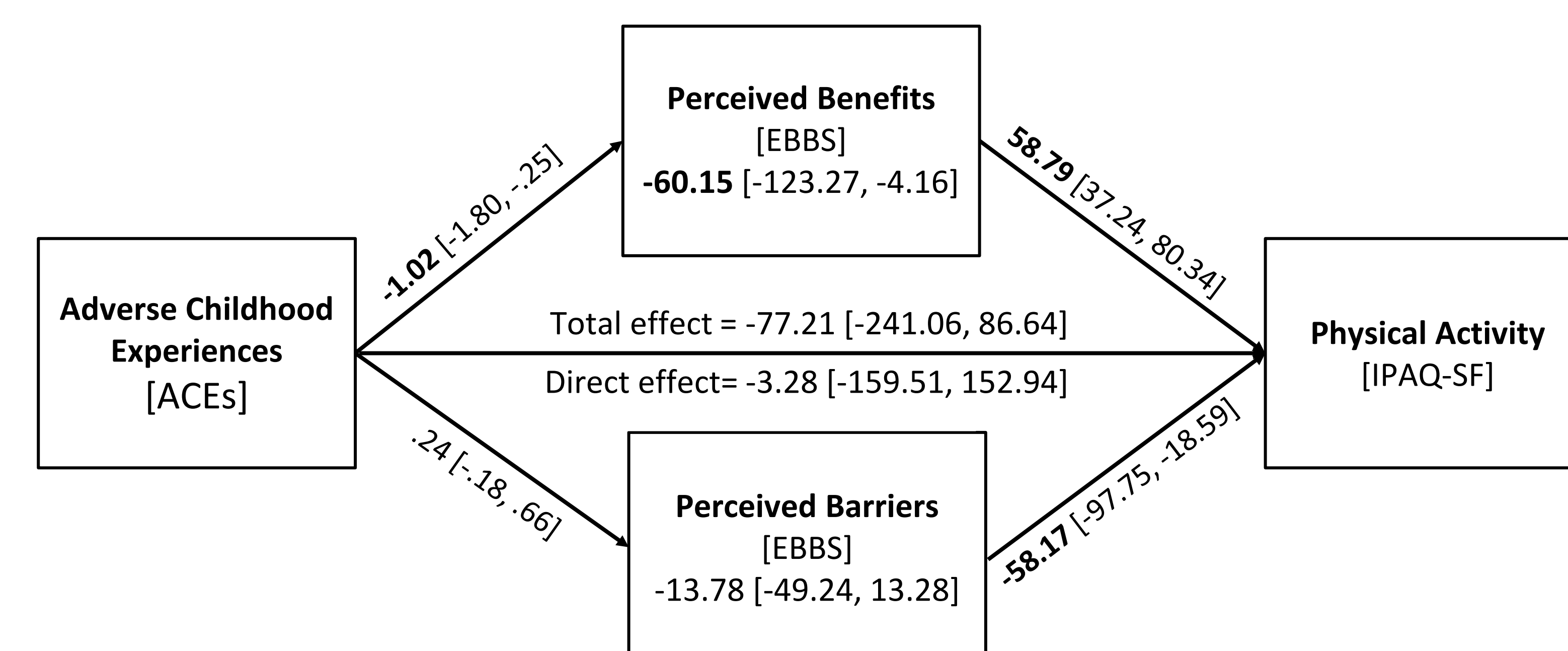


Figure 1. Parallel Mediation Model showing unstandardized effects and 95% CI. Bolded effect sizes indicate statistically significant effects ($p < .05$).

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